SMARTHACK XI · PPC "24H TO POWER PPC"

Prosumer Sentinel

One energy agent for Romania. It coordinates home batteries on each low-voltage feeder, watches wind turbines, and commits only what healthy, predictable assets can deliver.

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Midday solar is hitting feeder limits, and nobody can say what each asset should be doing

Network operations

- Rooftop export crosses the limit of an LV feeder at noon.
- The evening peak is imported hours later.
- Home batteries optimise alone, so absorption lands where no headroom is needed.

O&M

- Faults are found after the loss.
- Rules like "wind > 15 km/h and power < 10 kW" fire on every stop, curtailment and maintenance included.
- A broken PV string silently shrinks the flexibility a plan relies on.

Shared root cause: no model of what each asset should do given its weather.

Market size and PPC asset figures are unverified; see Assumptions.

The agent runs by itself from data to action, every 15 minutes

Perceive
Prosumer + wind SCADA, RO weather, OPCOM prices, GIS. Validated on the way in.

→

Estimate
Our models: PV and load P10/50/90, site health, wind normal behaviour, zone demand.

→

Qualify
Each asset gets one status: eligible, derated, excluded or human review.

→

Decide
Per-feeder LP, P10 commitment, O&M top-K per crew type.

→

Act
Battery schedules, commitments, work orders, alerts, park reports, GIS flags.

Every action is traceable

Audit row: model version, input hash, prediction, threshold, status. Realised vs committed is logged on the next tick.

One qualification contract links every model to every decision

A broken PV string lowers a feeder's committed MW. A repair returns capacity to the plan.

LIVE, 3 MINUTES

Demo

Baseline and Sentinel replay the same test week: identical weather, assets, limits and prices. Only the decisions differ.

Production → Demand → Feeders F7 → string failure at site 1042 → O&M park → results.

One model per problem, trained by us, each compared with a simple baseline

Model	Question	Architecture	Baselines	Tier
M1 PVNet	What should this PV produce?	PatchTST-style encoder + own reference window	pvlib, persistence, LightGBM	MUST
M2 LoadNet	What will this home consume?	Same encoder, separate weights	persistence, LightGBM	MUST
M4 SiteHealth	Is this site healthy?	1D-CNN on residuals + 14-day summary	threshold, CUSUM	MUST
M6 WindNBM	Is this turbine normal?	Small GRU + classifier on residuals, park context	WindMon rules	SHOULD
M3 DemandNet	Where will demand peak?	Aggregated M2 scenarios per zone	persistence	SHOULD
M8 GridTwin	Is the GIS feeder/phase right?	Voltage correlation + spectral clustering	GIS as-is	SHOULD

Data: Open-Meteo/ERA5, OPCOM and Energy-Charts for Romania; synthetic prosumers and turbines built on physics. Train 2022–23, validate 2024, test 2025, with 30% of sites and one wind park never seen in training. PVDAQ is non-RO validation. Every number says synthetic RO, real RO or non-RO.

Decisions come from real constraints, and we never sell the nominal number

Per-feeder plan (LP)

$$\Sigma\,(P90 - L10 - \text{charge} + \text{discharge}) + \text{rest} \le G + \text{slack}$$

$$e \ge \text{customer comfort reserve } R$$

$$G = \tau \cdot G_{max} + (1 - \tau) \cdot G_{conservative}$$

Worst plausible export. Low GIS trust only ever tightens the limit. When surplus exceeds headroom, we show the residual slack.

P10 commitment

$$\text{committed} = (1 - h) \cdot Q10(\text{deliverable})$$

$$\text{nominal} \ge \text{available} \ge \text{committed}$$

Conformal-calibrated scenarios replayed through the battery model. h is the telemetry-loss haircut. Realised is measured afterwards, misses published.

O&M triage: capacity is real

$$\text{score} = P(\text{fault}) \times (\text{lost kWh} \times \text{price} + \text{lost flex MWh} \times \text{value}) \times \text{days until next visit}$$

Top-K per day per crew. Sensor faults get a remote check, never a truck. Curtailment gets no work order.

Results on the separate test set, misses included

Claim	Baseline	Ours	Label	Run
Energy over feeder limit (week)	self-consumption 1397 kWh	**94.5 kWh (−93%)**	synthetic RO	replay#506
Minutes over limit (not zero)	6645	1905	synthetic RO	replay#506
P10 commitment hit rate (target 90%)	n/a	**92.0%**	synthetic RO	replay#506
Comfort breaches; saving	1165 (rule has no reserve logic)	0; 23.7 lei/home/month	synthetic RO, real prices	replay#506
PV nMAE, seen / unseen sites	LightGBM 0.041 / 0.045; pvlib 0.083 / 0.093	**0.015 / 0.036**	synthetic RO	eval#521
PV nMAE, real US systems	LightGBM 0.283; pvlib 0.312	0.268	non-RO (PVDAQ)	eval#521
Load nMAE, seen / unseen	LightGBM 0.256 / 0.284; persistence 0.361 / 0.390	0.246 / 0.285 (tie unseen)	synthetic RO	eval#521
Site faults PR-AUC, seen / unseen	threshold + CUSUM 0.49 / 0.40	**0.73 / 0.56**	synthetic RO	eval#521
Fault episodes found; median detect	40%; 35.9 h	**97.5%; 21.5 h**	synthetic RO	eval#521
Wind: recall; precision; curtail-vs-fault F1	WindMon 0.75; 0.84; 0.39	**0.89; 0.92; 0.93**	synthetic RO, held-out park	train#593/594
National demand nMAE, day ahead	persistence 0.067	**0.039**	real RO	make demand

Replay is 5 picked feeders with 46% batteries. A whole fleet cuts energy over limit 67% at 45% batteries, 30% at the real 20% (replay#583, #568), so value scales with adoption. The simulator is pvlib-based, so synthetic wins are partly circular. Tuned per-home rule not run.

Where the models are wrong, and what we tried

Where it is wrong

- **Forecast bands don't transfer.** PV 10–90 coverage: 0.88 seen, 0.61 unseen, 0.10 on PVDAQ, against a 0.78–0.82 target (eval#521).
- **Load:** only a tie with LightGBM on unseen sites, 0.285 vs 0.284 (eval#521).
- **Real PV faults:** PR-AUC 0.43 vs 0.32 on PVDAQ + injected (eval#521). Soiling is hardest, AP 0.29 (train#297).
- **Minutes vs energy:** F7 on Monday goes 285 → 375 minutes over while energy falls 79 → 27 kWh; the plan penalises energy (replay#506).
- **Demand peak hour:** error 0.30 h, no better than persistence at 0.29 h (make demand).

What we tried

- **Retrained on the shared fleet** after an independent check showed the first models lost to LightGBM (train#228 → eval#521).
- **P10 hit rate** rose from 82.2% to 92.0% with the fleet-trained load model (replay#380 → #506).
- **LP on negative prices:** fixed an unbounded case that sent 10 of 35 feeder-days to a greedy fallback; all 35 now solve (replay#322 → #380).
- **Telemetry quality gate:** corrupted units had dropped P10 to 0.23; the gate restores 0.97 (local drills on a mock world).

Robust by design: it can say "I'm not sure", and it shows why

Hard situations

- **Stale telemetry, sensor fault, data gap:** excluded and removed from the commitment.
- **Wide interval, ensemble disagreement, OOD:** human review, no truck.
- **Wrong GIS:** low trust tightens the limit.
- **A model slips:** its fallback runs, so a panel degrades, never the demo.

Why it decided

- Expected vs actual band for every site.
- Status with its reason on every asset.
- Which sites carried the absorption on a feeder.
- Audit trail from action back to the prediction.

Drills on the test set (eval#521): with 40% gaps PV nMAE is 0.058 vs LightGBM 0.096. Unit errors still break the forecast (0.76–1.15), so they are gated before any commitment.

What PPC would run, and what we assumed

Value

- PPC pays; prosumers opt in free, with a customer-chosen comfort reserve and override at any time.
- First pilot value: feeder congestion relief.
- Wind: fewer false alarms; we quote lost MWh and lei per event from simulation, not a fleet-wide saving.
- Balancing revenue is not assumed.

Assumptions, not verified

- Battery control is simulated; no hardware was touched.
- Demo feeders are hand-picked, with 46% batteries vs about 20% in Romania (ANRE, Nov 2025).
- Export remuneration β and compensation rules.
- Aggregator rules for small assets; distribution/supply data sharing; GDPR consent.
- Prosumer market figures and PPC wind fleet details.
- Battery cost and cycle life are inputs, not results.

Run it yourself

```
git clone github.com/Andrei9383/smarthack
make reproduce      # data → sim → train → eval → replay
npm ci && npm run build && npm start
```

Built in 24 h

Models, data pipeline, LP, agent and app, with the commit history to show it.

Prior work

Andrei's WindMon (PPC HackITall, April 2026): SCADA schema, quality and anomaly rules as our wind baseline, dashboard layout as a visual reference.

Data

Public or synthetic only, sources cited in the README. Results regenerate from the evaluation script.